

ARITHMETIC TOPOLOGY, EXECUTABLE

primeknots

Named theorems of number theory, verified from scratch in C++
— where the process is designed to catch fitting the answer.

github.com/DerekMarshall/primeknots · 63 tests, 0 failing · MIT

A survey that read as mathematics

- A research roadmap mixed **real** arithmetic topology with a **fabricated** object — a "chaotic operator O_x " with invented properties.
- In prose, the fiction was indistinguishable from the facts: same register, same citations-shaped scaffolding.
- Nothing in the reading experience separated the true claims from the false one.

The behavior, not the tool: **unverified synthesis** — assembling plausible text without a mechanism that could reject a wrong piece.

Why it survived five revisions

- Each critique was **absorbed as camouflage**: the text got more plausible, not more true.
- Every review pass raised the polish and lowered the visibility of the seam.
- Agreement with a reviewer is not evidence. Neither is agreement with expectation.

Related behaviors seen elsewhere in the project's own history: **fitting a result to an expected value**, and **fabricating a citation**. Same root — no falsifying step.

Rule 1

Agreement with expectation is evidence **only** when the process makes fitting **detectable and costly**.

Never fit conventions to expected answers. If a published anchor comes out wrong, the bug is in the code or in our *reading of the source* — go fix the reading. Flipping a sign until the known number appears is falsification of the experiment.

Everything after this slide is the machinery built to enforce Rule 1.

ACT II

The dictionary and the ladder

One analogy — primes as knots — climbed one theorem at a time, each rung a running test.

Primes $\leftrightarrow$ knots

- $\text{Spec } \mathbb{Z}[1/S] \leftrightarrow$ a 3-manifold with boundary (one "knot" per prime in S).
- Legendre symbol $\leftrightarrow$ linking number.
- Rédei symbol $\leftrightarrow$ triple (Milnor) linking.
- class group $\leftrightarrow H_1$; ζ -zeros $\leftrightarrow$ a spectrum.

Morishita's *Knots and Primes* [M12] is the dictionary; the whole ladder is this one analogy, made executable.

Stage 0 reciprocity
Stage 1 linking matrix
Stage 2 Rédei / Borromean
Stage 3 4-rank
Stage 4 Chern–Simons (mod 2)
Stage 5 explicit formula
Stage 6 S_3 Dijkgraaf–Witten

Reciprocity, the base case

- Legendre/Jacobi symbols, Tonelli–Shanks, Miller–Rabin — from scratch, u128 built-ins only.
- **Twin:** every symbol computed by Euler's criterion *and* by the reciprocity law — two algorithms, cross-checked.
- **Three coverages, stated separately** (not one "exhaustive to 10^7 "):
 - the *reciprocity law* over all pairs of odd primes $< 3 \times 10^4$ — 5,260,146 pairs;
 - both *supplements* over all 664,578 odd primes $< 10^7$;
 - the Euler-vs-reciprocity *twin* — 53,165,634 symbol agreements, exhaustive to 10^7 .

✓ verified · reciprocity 5,260,146 pairs to 3×10^4 · supplements 664,578 primes to 10^7 · twin 53,165,634 agreements to 10^7

The linking matrix

The linking matrix of primes

Stage 1 — $(p/q) = (-1)^{lk_2(p,q)}$ over primes $\equiv 1 \pmod{4}$. Renders emitted JSON only.

Linking heatmap



■ linked (symbol -1) ■ unlinked (symbol +1)

The matrix $(p/q) = (-1)^{lk_2(p,q)}$ over primes $\equiv 1 \pmod{4}$. It is **symmetric** — the picture *is* quadratic reciprocity.

✓ verified · symmetry over 39175×39175 (1.53×10^9 bits, no sampling)

The Rédei symbol

- The triple symbol $[a, b, c] \in \{\pm 1\}$ is the mod-2 triple **Milnor invariant** — a Massey product in Galois cohomology (Morishita [M04]).
- **Rédei reciprocity**: full S_3 symmetry under all six permutations of the arguments — 2022 mathematics (Stevenhagen [S22]), verified over every valid triple in range.
- The 2-adic normalization is the delicate part: [pinned to the source](#) before any code (stage2-pinning §8), with *invariance senior to anchor*.

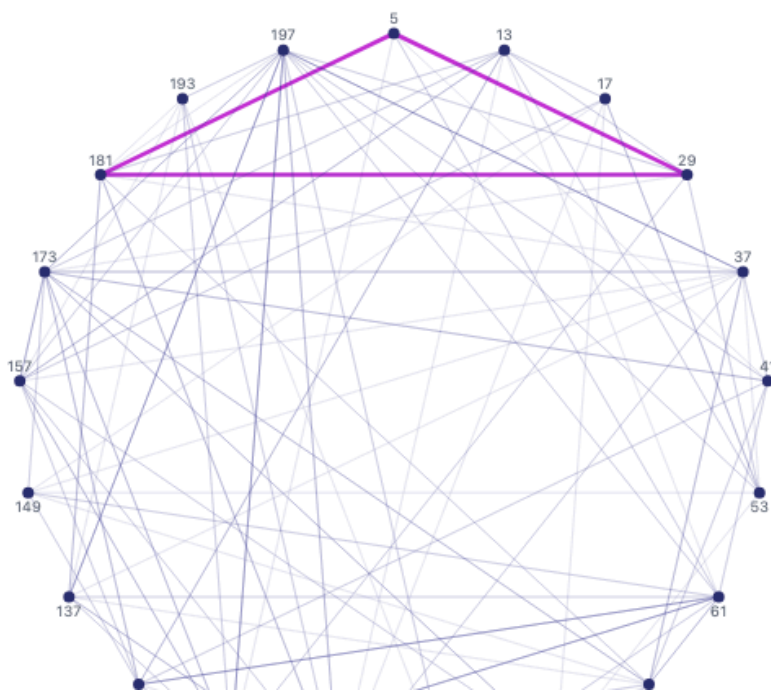
✓ verified · 702,788 triples × 6 = 4,216,728 symbol evaluations agree

Borromean primes

Borromean prime triples

Stage 2 — pairwise-unlinked triples of primes $\equiv 1 \pmod{4}$ with Rédei symbol -1 (triple-linked). Renders emitted JSON only.

3-uniform hypergraph



Triples pairwise *unlinked* yet triple-linked ($[a, b, c] = -1$). Anchor: $[13, 61, 937] = -1$, the standard Borromean triple (Morishita [M12]); the *normalization* is pinned to Stevenhagen [S22]/Corsman [C07].

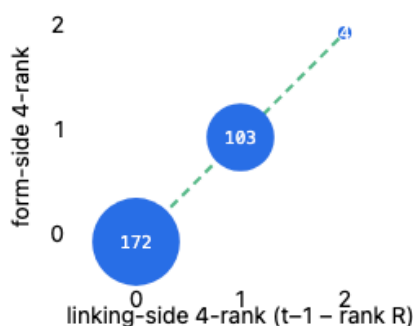
✓ verified · $[13, 61, 937] = -1$, the [M12] standard Borromean triple

Class groups as H_1 — the 4-rank

Class groups and the Rédei–Reichardt 4-rank

Stage 3 — narrow class group $Cl^*(D)$ computed two independent ways. The scatter must be a perfect diagonal. Renders emitted JSON only.

4-rank: form composition vs. linking matrix



each cell = # discriminants with (linking corank $t-1 - \text{rank } R$, form-side 4-rank); off-diagonal (red) = a Rédei–Reichardt disagreement

Aggregate

Rédei–Reichardt over 279 discriminants (primes $\equiv 1 \pmod{4}$, $t \leq 4$, $D \leq 20000$):

form-side 4-rank == linking-side 4-rank: **ALL MATCH ✓**

Rédei–Reichardt: the 4-rank of the narrow class group (**form** side) equals the F_2 -rank of the linking matrix (**linking** side). Two computations, one diagonal.

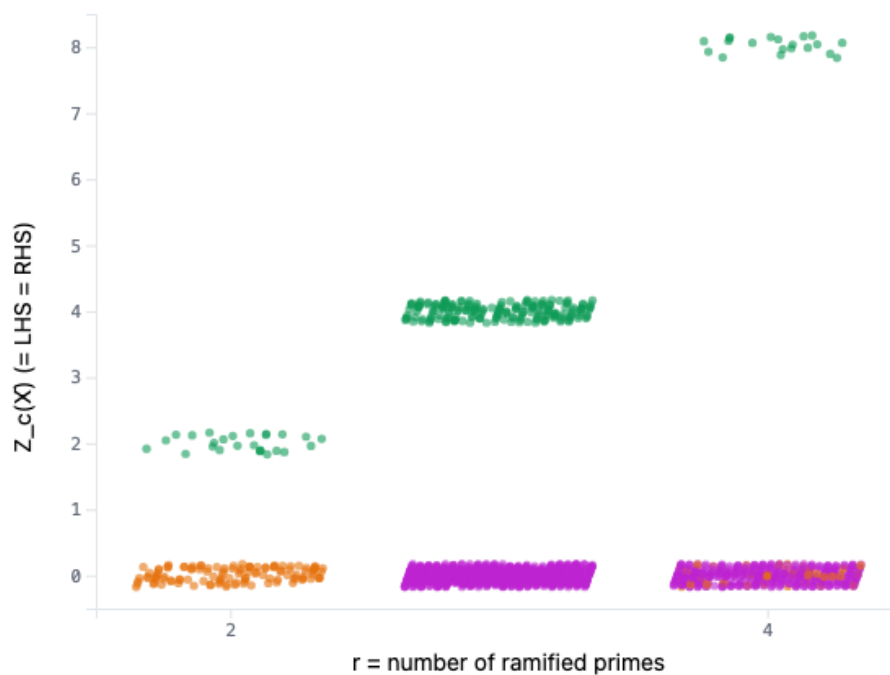
✓ verified · 1,986 discriminants; form-rank = linking-rank; oracle bnfnarrow

Arithmetic Chern–Simons (mod 2)

The arithmetic Chern–Simons partition function

Stage 4 — mod-2 Dijkgraaf–Witten $Z_c(X)$ for $K=\mathbb{Q}(\sqrt{(p_1\cdots p_r)})$, $p_i \equiv 1 \pmod{4}$ (Hirano route). LHS (brute character sum) vs RHS (closed form). Renders emitted JSON only.

Z_c by number of ramified primes and phase class



The mod-2 Dijkgraaf–Witten partition function Z_c , computed as a brute character sum (LHS) and a closed form (RHS) — in **separate translation units, disjoint authoring sessions**. Agreed on first comparison.

✓ verified · LHS = RHS over 386 gated prime sets (4,810 extended)

The degenerate case, named as such

The mod-2 (Hirano) case is the **degenerate** one: Chern–Simons is F_2 -linear, so the identity *reduces to linear algebra on the linking matrix*.

- The contribution was the **pinning** and a **hypothesis erratum**: RESEARCH.md's " $p = 2$ over $\mathbb{Q}$ " is unlicensed — CKKPPY Thm 1.1 needs odd p over a totally imaginary field.
- The odd- p Gauss-sum case is **future work**, not a result here.

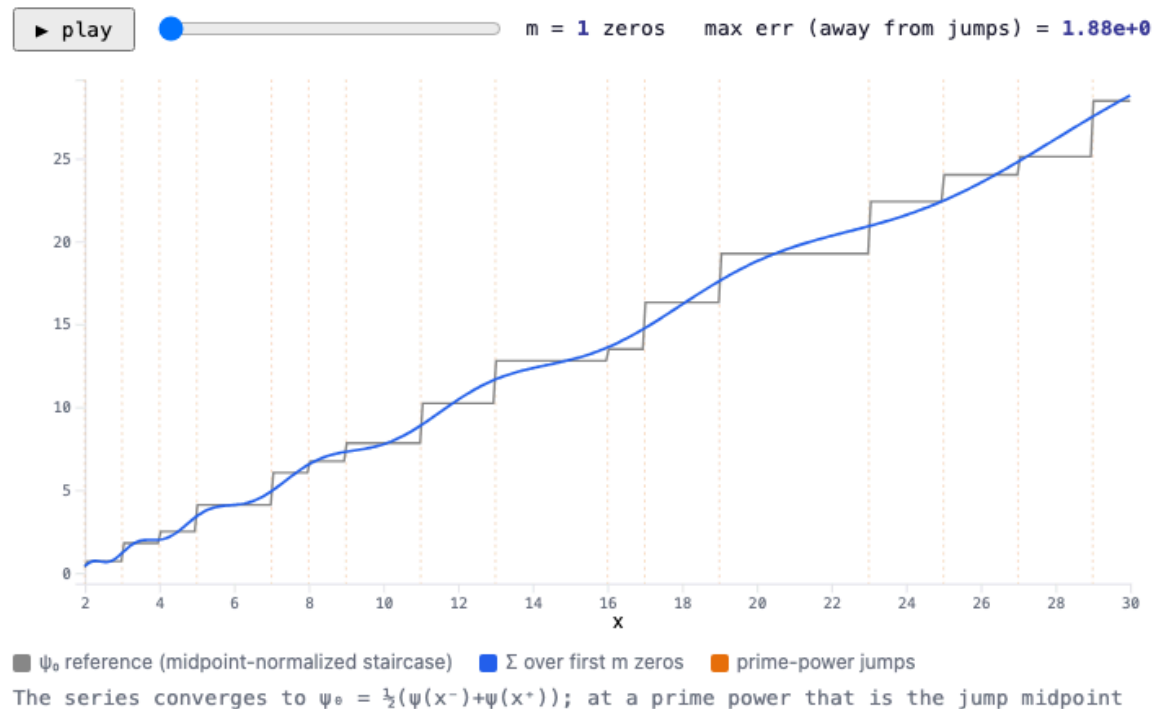
A slide that oversold this would fail our own review.

The Weil explicit formula

The explicit formula, and the Deninger gap made visible

Stage 5 — primes $\Leftrightarrow$ zeta zeros (Weil/von Mangoldt explicit formula), beside a fully-constructed dynamical system where the same identity is geometry. Renders emitted JSON only.

$\psi_0(x)$ reconstructed from ζ zeros — the spectral side building the prime staircase



$\psi(x)$ rebuilt from ζ zeros. Zeros by Riemann–Siegel, twinned against Euler–Maclaurin, completeness **Turing-certified** (not Gram's law). Beside it: a model flow (next slide).

✓ verified · first 10^4 zeros match Odlyzko to ≥ 8 dp (max dev 2.9×10^{-9})

✓ verified · converges to the jump midpoint ψ_0 , not $\psi(x)$

Where the identity IS geometry — and where it isn't

- A subshift's Ruelle zeta: **Euler product over closed orbits = $1/\det(I-L_s)$** , exactly — orbits genuinely exist, the identity is geometry.
- Arithmetic satisfies the *same identity shape* (primes $\leftrightarrow$ orbits, zeros $\leftrightarrow$ spectrum)...
- ...but **no such dynamical system is constructed.**

Deninger's foliated flow:
not constructed — that gap is the program [D98].

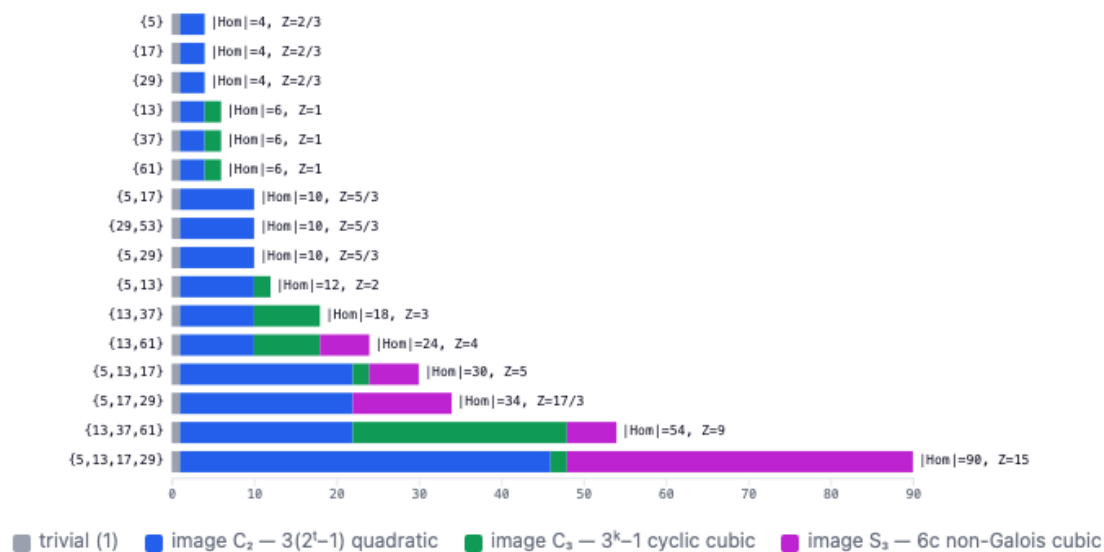
✓ verified · Euler product = determinant, agreement $\sim 9 \times 10^{-16}$ over an s-grid

Non-abelian S_3 counting

The S_3 Dijkgraaf–Witten invariant of $\text{Spec } \mathbb{Z}[1/S]$

Stage 6 — $Z_{\text{DW}} = |\text{Hom}(\pi_1^{\text{et}}, S_3)|/6$, decomposed by image: $1 + 3(2^t-1) + (3^k-1) + 6c$. The trivial/ C_2 / C_3 terms are our own machinery; the S_3 term $6c$ is refereed by cubic-field enumeration. Renders emitted JSON only.

$|\text{Hom}(\pi_1, S_3)|$ decomposed by image, per prime set S



N_{S_3} (= c, cubic iso classes) vs number of ramified primes t

$Z_{\text{DW}} = |\text{Hom}(\pi_1, S_3)|/6 = 1/6 + (2^t-1)/2 + (3^k-1)/6 + c$. The S_3 term c (cubic iso classes) refereed **two independent ways**: PARI nflist and K. Belabas's cubic.

✓ verified · mass formula exact over 16 sweep instances (exact rational)

✓ verified · Belabas = PARI on all 14 feasible sets (2 skipped: $D_{\text{max}} = \lceil p^2 \rceil > \text{the } 10^7 \text{ full-enumeration bound}$)

ACT III

The harness

The part that makes agreement mean something.

How every claim is tested

- `twin_` — two independent algorithms agree (never optimize both; keep the naive one as referee).
- `theorem_` — a named theorem over an exhaustive/sampled range.
- `anchor_` — a specific published value (e.g. $[13,61,937]=-1$).
- `invariance_` — independence from an arbitrary choice (sqrt branch, solution, ordering).
- `oracle_` — PARI/GP, the Odlyzko table, Belabas's cubic — referee, never in `src/`; SKIP *visibly* when absent.

`invariance_` is senior to `anchor_`. A passing anchor with a failing invariance means the convention is still wrong.

Gates, pinning, pre-registration

- **Two-phase gates:** convention-free code first; the normalization is *pinned to a source* in docs/notes/ before the load-bearing code is written.
- **Pre-registration:** Stage 4's LHS and RHS were built in separate translation units, in disjoint sessions, each seeing only its own spec. A mismatch would have been a *deliverable*, not a bug to reconcile.
- **Stratify the sweep — and sometimes prove the stratum void:** mod-8 third slot, corank cells, all-odd $\times$ odd-r (empty by parity), and the signature stratum void by Stickelberger.

Scope: this is $n=1$ — one project, one problem class, one operator. It is an existence proof that the discipline can be applied here, not evidence that it generalizes.

ERRATA — everyone was caught

Every party that touched this project was caught in at least one error. Every catch was a [computation](#) or a [citation](#) — never an argument from authority.

```
roadmap generator — 0_x, by a coherence check
spec author — two RESEARCH.md errata, by pinning + a witness
value
human reviewer — R1, equal-parity (killed by the √65 witness),
signature-mix (void by Stickelberger)
external (LLM) referee — a fabricated citation, by the verbatim-
quote protocol — the same referee made a genuine catch
coding agent — strat counter (parity), signed-a (associativity),
generator cap (sweep), 2c/6c (mass formula), PDFs-in-history,
missing headers (a second compiler)
```

(A selection — the full ledger has 18 rows, including this deck review's own two and the CI freshness guard's finding.)

```
✓ verified · 18 entries in docs/ERRATA.md (selection above); authority-
catches: 0
```

ACT IV

Results and horizons

What was verified

- The full ladder, Stages 0–6, green.
- Every number on these slides is **verified numerically over N cases**, N pulled from emitted JSON or test output — never "proved" (that word is for literature theorems, with citation).
- Empirics are labeled **observed**, not asserted: Borromean rate 51.6%; negative-Pell fraction 0.666 vs the Koymans–Pagano asymptotic 0.5805; $N_{\{S_3\}}$ growth.

✓ verified · 63 tests, 0 failing (10 oracle-refereed); extended run all green

What is NOT claimed

- **No new mathematics.** Every theorem verified here is known (Gauss 1801 through Stevenhagen 2022); the outputs are the verification, the tables, and the method.
- Verified numerically over N cases $\neq$ **proved**. No slide asserts anything the suite doesn't test.
- Deninger's dynamical system is **not constructed** — the gap is the program, not a result.
- The mod-2 Chern–Simons case is **degenerate** (F_2 -linear); the odd- p Gauss sum is future work.

Where it goes

- **Odd-p arithmetic Chern–Simons** over totally imaginary fields (CKKPPY in-hypothesis) — the non-degenerate case, as future infrastructure.
- **Murmurations** — an active area (2022): statistical structure in arithmetic data that this kind of harness is built to probe.

References: RESEARCH.md §10. The pictures on these slides are the repo's own viewers, rendering the same emitted JSON the tests consume.